

Taylor Nanjala Wanyama

wanyamataylor64@gmail.com | 0796 410 764 | [linkedin.com/in/taylor-wanyama-421920271/](https://www.linkedin.com/in/taylor-wanyama-421920271/)
github.com/taylorwanyama

Professional Summary

A highly motivated and solutions-oriented Data Scientist and AI/ML Engineer with a strong background in Financial Engineering. I specialize in designing and implementing robust machine learning and deep learning models to deliver data-driven, research-based solutions that optimize performance and mitigate risk.

Education

JKUAT, Bachelor of Science in Financial Engineering

Sep 2021 – May 2025

- Grade: Second Class Upper Division
- **Coursework:** Financial Modeling, Portfolio Management, Risk Analytics, Statistics, Insurance, Python, SQL, R

Experience

Junior Data Analyst, Onja Uone Ltd

Apr 2024 – March 2025

- Streamlined data collection and reporting workflows, reducing processing time by 20% and improving operational efficiency.
- Automated Python data cleaning workflows, reducing reporting lag by 40%.
- Built Tableau dashboards used by 5+ executives, improving KPI visibility and self-service analytics.
- Led A/B tests that reduced defect reports by 12% and increased user retention by 8%.
- Conducted market research and trend analysis, identifying 10+ actionable insights that informed key strategic decisions.
- Collaborated with 2+ cross-functional teams to gather requirements, define project scope, and ensure alignment with business goals.
- Delivered 15+ reports and presentations, communicating insights and recommendations that influenced decision-making.

Projects

PCOS Diagnostic Agent

github.com/taylorwanyama

- Trained a Random Forest classifier on 600+ medical records, achieving 91 % accuracy and 0.95 ROC-AUC, improving the early detection accuracy of PCOS over baseline models.
- Developed an AI agent with LangGraph and LangChain and a Groq LLM, enabling real-time diagnostic predictions, explanations, and treatment recommendations for clinicians.
- Implemented a logging system using SQLite, tracking predictions and explanations for continuous monitoring.
- Published the entire solution (pipeline, interface, documentation) on GitHub to ensure reproducibility and transparency.

E-commerce Product Recommendation System (Kilimall Beauty & Health)

github.com/taylorwanyama

- Built an end-to-end recommendation engine for 700+ beauty & health products, increasing product discovery efficiency by 30% in offline tests.
- Engineered a scraping pipeline (Selenium, BeautifulSoup, requests) that automated data collection from dynamic Kilimall pages, eliminating manual data entry.
- Integrated TF-IDF vectorization and cosine similarity to provide relevant recommendations with 80% precision, improving the browsing experience.

Digital Sales Dashboard (Tableau)

github.com/taylorwanyama

- Automated data preprocessing pipeline for 10,000+ records.
- Cut cleaning time by 70% using robust validation and outlier detection.

- Documented pipeline and published on GitHub.

Digital Sales Dashboard (Tableau)

github.com/taylorwanyama

- Processed 1,000+ transaction records, reducing cleaning time by 70% with automated preprocessing in Python.
- Built a real-time Tableau dashboard for sales and customer sentiment data, used by managers to track revenue by region, product performance, and churn signals.
- Identified 5+ underperforming products and presented insights that influenced product portfolio adjustments.
- Published the interactive dashboard on Tableau Public, allowing easy stakeholder access and self-service analysis.

Certifications

Machine Learning with Python – IBM (2025)

- Built and evaluated supervised & unsupervised machine learning models using scikit-learn.
- Applied preprocessing, model training, and performance evaluation on datasets.

Deep Learning Specialization – Cognitive.AI (2025)

- Designed CNN, RNN, and deep architectures for image and sequence data.
- Implemented models using TensorFlow and Keras, focusing on optimization.

Intermediate Machine Learning – Kaggle (2024)

- Developed skills in handling missing values, feature engineering, and model tuning.
- Worked with advanced techniques like random forests, gradient boosting, and cross-validation on real datasets.

Data Analysis with Python – IBM (2024)

- mastered data wrangling, visualization, and statistical analysis using Python (Pandas, Matplotlib, Seaborn).
- Applied exploratory data analysis (EDA) techniques to uncover insights from structured datasets.

Python for Data Science – IBM (2024)

- Learned to use Python for data science workflows: data collection, cleaning, transformation, and visualization.
- Practiced using libraries like NumPy, Pandas, and Matplotlib for basic analytics and reporting.

Python Essentials 1 & 2 – Cisco Networking Academy (2023)

- Built a strong foundation in Python programming, covering control flow, functions, modules, and OOP.
- Gained practical problem-solving experience through coding exercises and algorithmic thinking.

Technical Skills

Languages: Python, R, SQL

Data Tools: Pandas, NumPy, Excel, Tableau, Power BI, Plotly, Dash

ML/AI: Scikit-learn, TensorFlow, PyTorch, Keras, LangGraph, LangChain, Deep Learning, NLP

Databases: MySQL, **Tools:** Jupyter, RStudio, VS Code, Git, LaTeX, GitHub, Microsoft SQL Server

Concepts: Forecasting, Predictive Modeling, Risk Analysis, Scenario Analysis, Portfolio Optimization, Credit Scoring

Key Areas of Interest

Financial Modeling and Valuation, Risk Management, Machine Learning in Healthcare and Finance, Time Series Forecasting, AI-Driven Systems

References

- **Lec. Dennis Ogot**

Lecturer, Financial Engineering, JKUAT

Email: ogot.dennis@jkuat.ac.ke | Phone: +254-705-979-402

- **Augustine Wekesa**

Sales Officer, Onja Uone Ltd

Email: augustinewasilwa543@gmail.com | Phone: +254-727-489-240